NEPHROLOGY TEACHING SERIES

Chronic Kidney Disease

Outpatient management — the four pillars

Dialysis initiation is driven by symptoms and complications, not by eGFR alone.

Dr. Cheng • Premier Nephrology Medical Group

MS3/MS4 rotation teaching

Why it matters

CKD is common, silent, and modifiable — most of the work is outpatient.

WHAT THE CLINIC VISIT COVERS

- Stage + cause — eGFR, UACR, trend
- Four pillars: ACEi/ARB, SGLT2i, finerenone, GLP-1 RA
- BP, volume, K, bicarb
- Anemia + CKD-MBD screening
- CV risk: statin, lifestyle
- Vaccinate: flu, COVID, pneumo, HBV
- Modality education by eGFR < 30; transplant referral as eGFR approaches 20

TEACHING PEARL

Never chart just an eGFR. CKD is CAUSE + GFR STAGE + ALBUMINURIA. eGFR 55 / UACR 800 beats eGFR 40 / UACR 10 — the risk drives the plan.

KDIGO 2024 Clinical Practice Guideline for Evaluation and Management of CKD.

KDIGO: Cause, GFR stage, Albuminuria (CGA)

Every CKD patient gets characterized on three axes.

	A1 (UACR < 30)	A2 (UACR 30–300)	A3 (UACR > 300)
G1 (≥ 90)	Low	Moderate	High
G2 (60–89)	Low	Moderate	High
G3a (45–59)	Moderate	High	Very high
G3b (30–44)	High	Very high	Very high
G4 (15–29)	Very high	Very high	Very high
G5 (< 15)	Very high	Very high	Very high

Risk categories drive visit cadence, referral, and intensity of therapy. Very-high-risk patients need nephrology follow-up every 3–4 months minimum.

KDIGO 2024 CKD Guideline. Use CKD-EPI 2021 (race-free) for eGFR.

The four pillars of renoprotection

For diabetic kidney disease, aim for all four when eligible; non-diabetic CKD uses selected pillars.

ACEi / ARB

First-line for UACR > 30
Max tolerated dose
Accept Cr ↑ up to 30%
Hold if K > 5.5
RENAAL, IDNT, Captopril, AASK

SGLT2 INHIBITOR

Dapa or empa 10 mg daily
Start at eGFR ≥ 20; continue to KRT
Benefit independent of DM
Expect eGFR dip ~4 mL/min
DAPA-CKD, EMPA-KIDNEY, CREDENCE

FINERENONE

Non-steroidal MRA; T2DM + albuminuric CKD
10 mg if eGFR 25–<60; 20 mg if ≥60
K+ ≤ 5.0 at start
Additive to ACEi/ARB + SGLT2i
FIDELIO, FIGARO

GLP-1 RA

Semaglutide 1.0 mg SQ weekly
T2DM + CKD with CV/kidney risk
Weight loss bonus
FLOW (2024) — 24% RRR

KDIGO 2024 CKD + 2022 Diabetes in CKD guidelines. DAPA-CKD, EMPA-KIDNEY, FIDELIO, FLOW.

Must-ask history and must-do labs

The backbone of every CKD visit.

MUST-ASK HISTORY

- Cause: DM, HTN, GN, PKD, stones, obstruction
- Home BP log (AM + PM readings)
- Weight changes / edema / dyspnea on exertion
- Urinary symptoms: frothy, hematuria, nocturia
- OTC and supplements: NSAIDs, PPIs, herbal
- Diet: protein load, Na+, K+ (bananas, potatoes), salt substitutes
- Smoking / alcohol — modifiable risk factors
- Prior kidney imaging or biopsy

LABS AT EACH VISIT (TIERED)

- **Every visit: BMP, UACR**
- **Q3–6 mo (stage 3+): CBC, Mg, PO4, bicarb**
- **Q6–12 mo (stage 3b+): PTH, 25-OH vit D, ferritin / iron sat, albumin**
- **Annually: lipid panel, HbA1c if DM**
- **Once: hepatitis serologies (transplant prep), HIV, renal US if never done**
- Home BP — bring log; if not, teach how to log

Blood pressure in CKD

KDIGO 2021: standardized office SBP < 120 mmHg when tolerated; routine/home targets are individualized.

MEASUREMENT MATTERS

- Use attended automated office BP (AOBP) when possible
- Home BP: AM + PM, 2 readings each, 1 min apart
- Correct cuff size; arm at heart level; quiet 5 min
- White-coat HTN is real — home/ambulatory BP catches it
- Masked HTN: normal in clinic, elevated at home
- **'120' target was derived from AOBP in SPRINT — not routine office BP**

DRUG SEQUENCE

1. ACEi or ARB (never both)
2. Thiazide or loop diuretic (loop if eGFR < 30)
3. Amlodipine or other CCB
4. Spironolactone or finerenone (resistant HTN)
5. Additional: beta-blocker, hydralazine, clonidine

Avoid: dual RAAS blockade (ONTARGET — harmful)

In DKD: favor ACEi/ARB + SGLT2i + finerenone combo

KDIGO 2021 BP in CKD Guideline. SPRINT trial (NEJM 2015).

Complications — screen and treat

The reasons patients feel bad and the reasons they die.

Complication	Screen	Treat
Anemia	Hgb, ferritin, TSAT (q6–12 mo stage 3+)	Iron first (TSAT < 30 or ferritin < 100). erythropoiesis-stimulating agent (ESA) if Hgb < 10 after iron (target 10–11.5). Daprodustat (HIF-PHI) emerging.
CKD-MBD	Ca, PO4, PTH, 25-OH D (q6–12 mo stage 3b+)	Treat persistent hyperphosphatemia with diet ± binders. Cholecalciferol for low 25-OH D. Reserve calcitriol/paricalcitol for severe progressive 2° HPT in G4–G5.
Metabolic acidosis	Serum bicarb (q visit stage 3b+)	KDIGO 2024: consider oral NaHCO3 when bicarb < 18. BiCARB trial did not confirm routine benefit.
Hyperkalemia	BMP q visit; symptoms; diet hx	Diet first. Binder (SZC, patiromer) lets you keep RAAS (AMBER). Avoid Kayexalate.
CVD risk	Lipid panel, HbA1c, smoking	Statin in most CKD (SHARP — not in dialysis). ASA if ASCVD. BP to target. Smoking cessation.

KDIGO 2017 CKD-MBD update. KDIGO 2012 Anemia of CKD (2026 update pending).

Preparing for kidney failure

Start the conversation at eGFR 30. Refer to transplant at eGFR 20.

MODALITY EDUCATION (eGFR < 30)

- Discuss: in-center HD, home HD, PD, transplant, conservative care
- Access planning at eGFR < 20: arteriovenous fistula (AVF) cannulation minimum 6–8 weeks, but plan ≥ 6 months ahead (maturation + possible salvage)
- Protect non-dominant arm veins — no PICC, no IVs from now
- Patient preferences, caregiver support, home environment
- Social work + dietitian involvement
- **Dialysis is not an eGFR threshold — it's a symptom threshold**

TRANSPLANT EVALUATION

- Early referral at eGFR ≤ 20 — allows preemptive listing
- Pre-emptive transplant = best outcomes
- Living donor > deceased donor (longer graft life)
- Workup: cardiac, cancer screen age-appropriate, HLA typing, HBV/HCV/HIV
- Dental clearance, vaccines UTD pre-transplant
- Barriers to screen: SES, adherence concerns, active cancer (usually 2–5 y wait)

KDIGO 2020 Transplant Candidate Evaluation. AST/ASN modality choice guidance.

Landmark CKD trials you must know

Organized by pillar.

Trial	Year	Takeaway
RENAAL	2001	Losartan ↓ ESKD/doubling Cr 25–28% in T2DM nephropathy.
IDNT	2001	Irbesartan — renoprotection independent of BP lowering.
ONTARGET	2008	Dual RAAS (ACEi+ARB) → harm, more hyperK and AKI.
SPRINT	2015	Intensive SBP < 120 ↓ CV events 25% (non-diabetic).
CREDENCE	2019	Canagliflozin ↓ kidney failure 30% in DKD.
DAPA-CKD	2020	Dapagliflozin ↓ kidney failure 39% (incl. non-DM).
EMPA-KIDNEY	2023	Empagliflozin ↓ CKD progression 28% across broad eGFR.
FIDELIO-DKD	2020	Finerenone ↓ CKD progression 18% in T2DM on max RAAS.
FLOW	2024	Semaglutide ↓ kidney disease progression 24% in T2DM + CKD.

Full citations on references slide.

Clinical case

Think it through — answer is on the next slide.

VIGNETTE

A 58-year-old Black man with T2DM × 15 yr, HTN, obesity. Labs: eGFR 38 (CKD-EPI 2021), UACR 720 mg/g, HbA1c 7.8%, K+ 4.8, HCO3 23, Hb 10.9 (ferritin 90, TSAT 18%), home BP avg 148/88. On lisinopril 40, amlodipine 10, metformin 1000 BID, atorvastatin.

QUESTION: Which pillars are missing, and what's your next 3 interventions?

Clinical case — answer

Key teaching points from the case.

ANSWER

Missing: SGLT2i, finerenone, GLP-1 RA. Also anemia workup needed. (1) Add dapagliflozin 10 mg, (2) add finerenone 10 mg if K+ remains acceptable, (3) IV iron then consider ESA if Hb stays < 10.

TEACHING

CGA classification: G3b-A3 (very high risk). DKD pillars should be on-board when eligible. Stop metformin at eGFR < 30; he's still OK at 38. BP target is standardized office SBP < 120 when tolerated; routine/home readings are often managed closer to <130/80. Also: retinal exam annually, vaccinate (flu, COVID, pneumococcal, HBV), modality education at eGFR < 30, nephrology follow-up q3-4 months for very-high-risk.

REFERENCES

Chronic Kidney Disease

Guidelines

KDIGO 2024 CKD Evaluation & Management

KDIGO 2022 Diabetes in CKD

KDIGO 2021 Blood Pressure in CKD

KDIGO 2017 CKD-MBD update

KDIGO 2020 Transplant Candidate Evaluation

Landmark Trials

RENAAL — NEJM 2001

IDNT — NEJM 2001

ONTARGET — NEJM 2008

SPRINT — NEJM 2015

CREDENCE — NEJM 2019

DAPA-CKD — NEJM 2020

EMPA-KIDNEY — NEJM 2023

FIDELIO-DKD — NEJM 2020

FLOW — NEJM 2024

SHARP — Lancet 2011

UpToDate (2026 access):

Overview of the management of chronic kidney disease in adults • *Proteinuria and albuminuria: evaluation and causes* • *Antihypertensive therapy and progression of CKD* • *SGLT2 inhibitors in CKD* • *Dialysis modality and vascular access planning*